DEVOPS CAPSTONE

CampusDesk

Campus IT, organized.

A helpdesk for campus requests, delivered through a tested container and Kubernetes workflow.

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APPLICATION

Every request has a clear next step

Students report network, hardware, software and account issues.

The support team sets priority, assigns an owner and resolves tickets.

Search and status filters make the queue easier to manage.

Synthetic demo tickets exercise the same APIs as newly created requests.

TECHNOLOGY

One application, three core layers

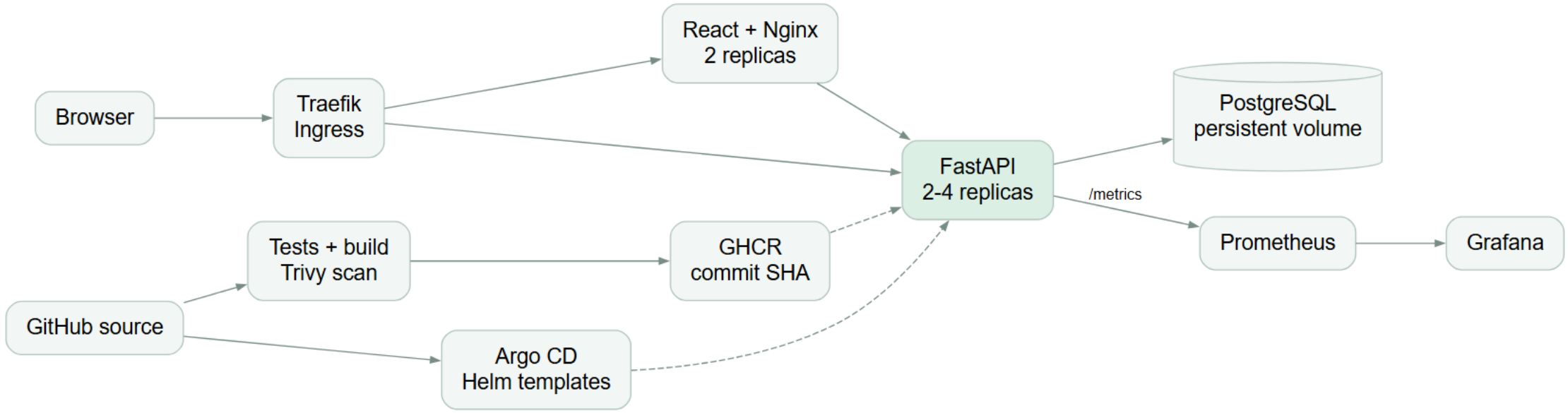
React + Vite + Nginx deliver the responsive service desk.

FastAPI validates requests; SQLAlchemy manages database operations.

PostgreSQL persists tickets; Alembic versions the schema.

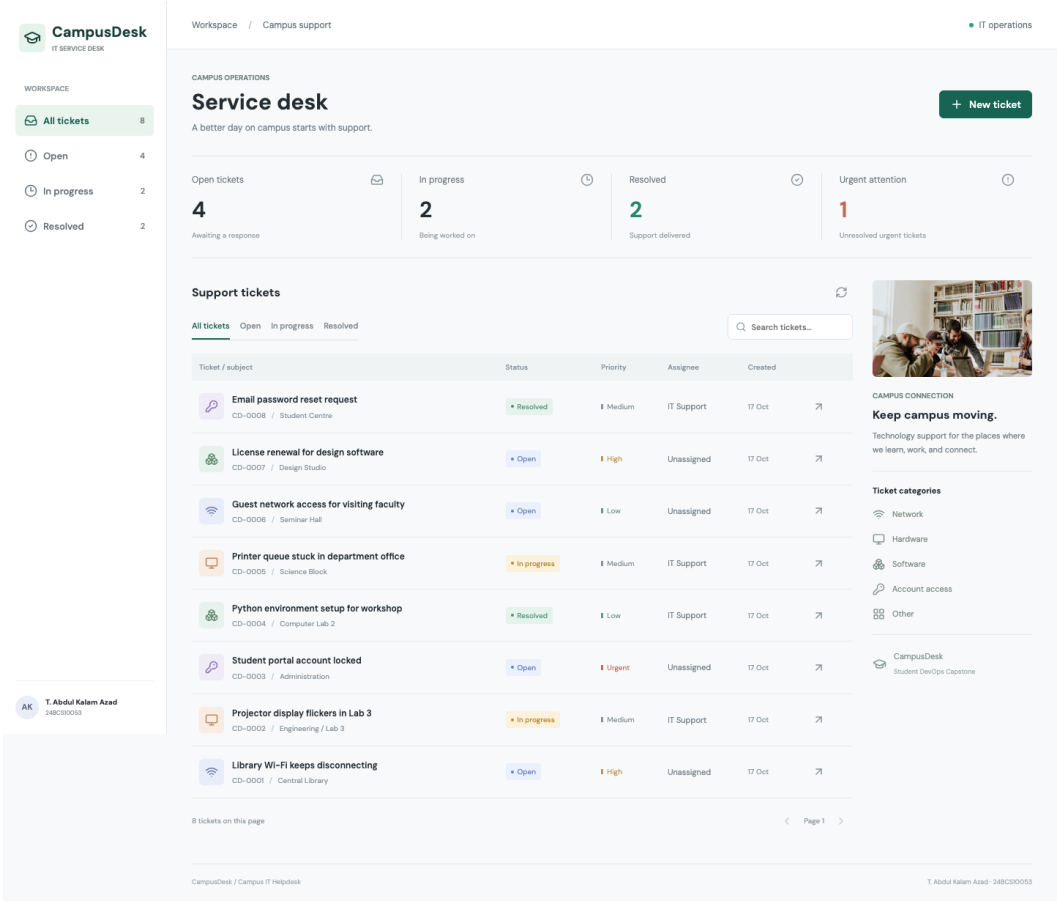
Python tests and Ruff catch regressions before image publication.

Traffic and changes follow defined paths



WORKING APPLICATION

The service desk runs in the browser



Quality gates precede publication

40 API tests, migrations, Ruff and source-security gates run first.

Trivy blocks HIGH and CRITICAL findings in both application images.

Native AMD64 and ARM64 jobs verify Helm and 17 browser checks each.

Only verified images become the multi-platform commit-SHA release.

Kubernetes manages application replicas

Helm defines Deployments, Services, Secrets and PostgreSQL storage.

Traefik routes the UI and API through one Ingress hostname.

Readiness checks database access; liveness checks the process.

Live load scaled the API from two replicas to four: 31,274 successful requests.

Git records the desired deployment

Argo CD reads the CampusDesk Helm chart from the GitHub repository.

Sync compares the committed configuration with the running cluster.

Migration ordering prepares the database before application rollout.

CI promotes scanned image tags; automatic sync deploys that Git revision.

Metrics explain what the service is doing

Prometheus discovers and scrapes each backend replica.

Grafana shows request rate, p95 latency, status codes and availability.

A bounded load generator creates real API traffic for the dashboard.

Backend availability and error-ratio alert rules detect service problems.

AWS infrastructure runs from code

Two public subnets and two private subnets span availability zones.

A NAT gateway connects private worker nodes to external registries.

EKS 1.35 and two private t3.medium workers are verified in Mumbai.

Plan: 61 resources. Apply recovered from DNS failure; the follow-up plan is clean.

Failures become evidence for diagnosis

A nonexistent image tag demonstrates ImagePullBackOff and pod events.

A wrong Service selector demonstrates missing endpoints.

A failed database connection makes readiness return HTTP 503.

Image scans guide remediation; successful deployment alone is not security proof.

Verification is part of the deliverable

A working app needs tests, repeatable deployment and observable behavior.

Screenshots and reports should show actual runs and identify their environment.

Delete load balancers and database volumes before Terraform destroys EKS.

The evidence index records results, limitations and the final cleanup checks.

Brief, rubric and submission guide: github.com/Nency-Ravaliya/devsecops-python